Lab 1

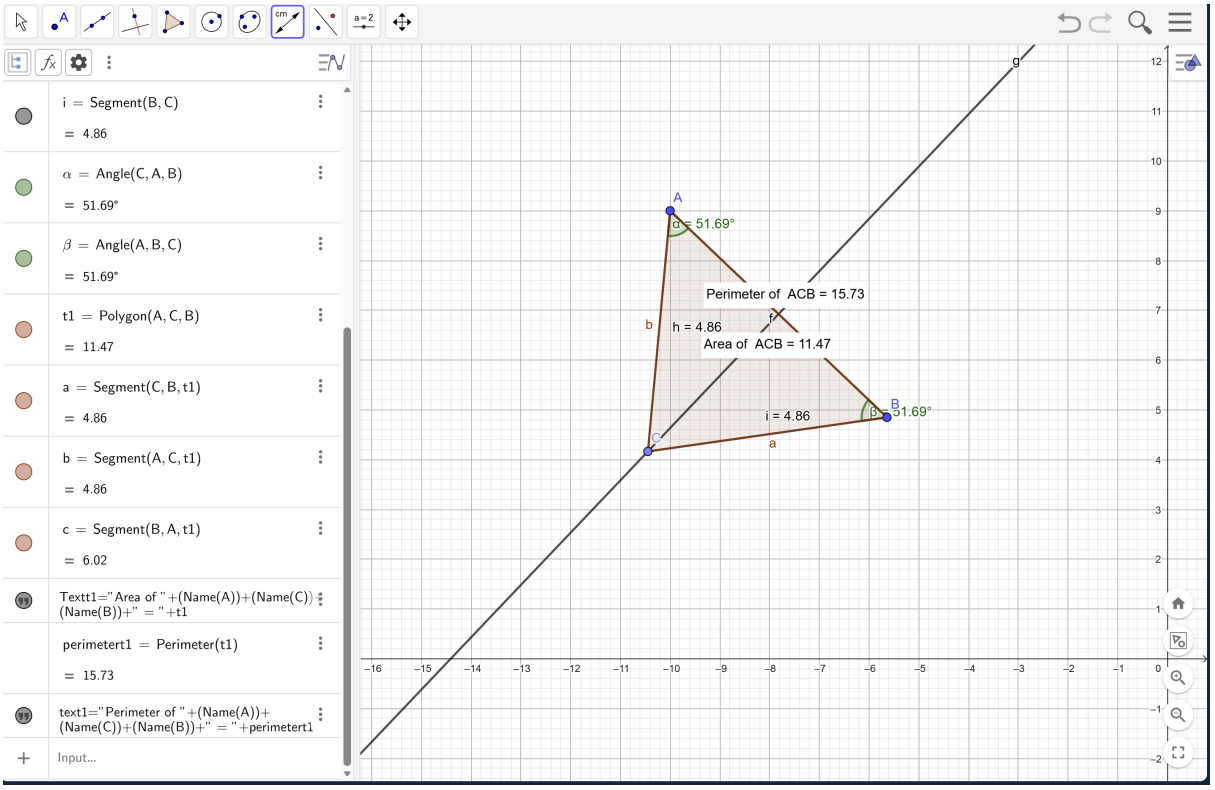
Created	@February 25, 2026 12:30 PM
Type	Laboratory
Reviewed	☐

Details

Most of the classes will be done on GeoGebra. Minimum grade of 5 to pass. 10 labs minimum for attendance.
We need an account on GeoGebra Classic

Task 1

- Start from a base segment and think about the geometric property that defines an
  isosceles triangle: the vertex must be equidistant from the endpoints of the base.
- Use a construction that guarantees equal distances instead of placing the third point
  manually.
- After constructing the triangle, measure the side lengths, angles, perimeter, and area.
  Verify correctness using the drag test.

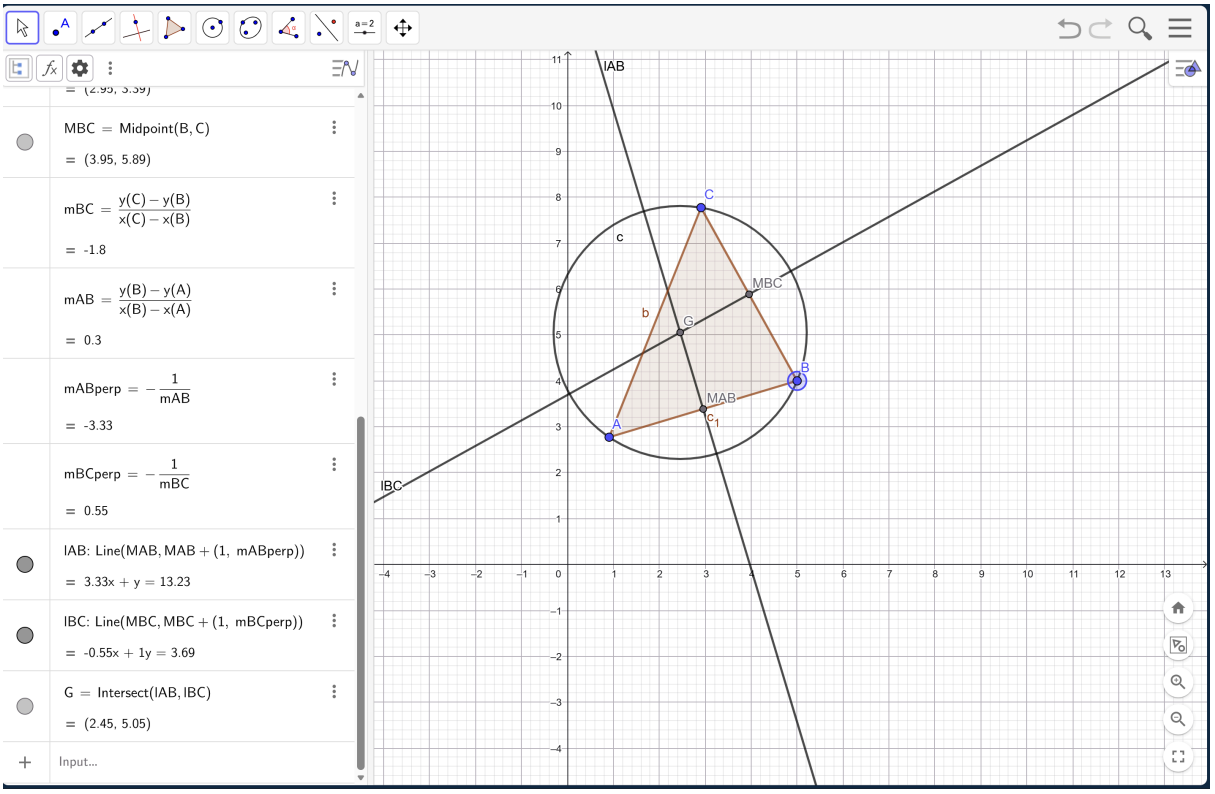


All sides have been verified and I also performed the Drag Test by dragging the triangle around. It still remains an isosceles triangle.

Task 2

Task 2 (15 min): Circumcenter & circumcircle; verify using the drag test.

- Remember that the circumcenter is the point that is equidistant from all three
  vertices of a triangle.
- Construct the geometric loci of points equidistant from two vertices and find their
  intersection. Use that point to define the circumcircle.
- Perform the drag test to confirm the circle always passes through all three vertices.

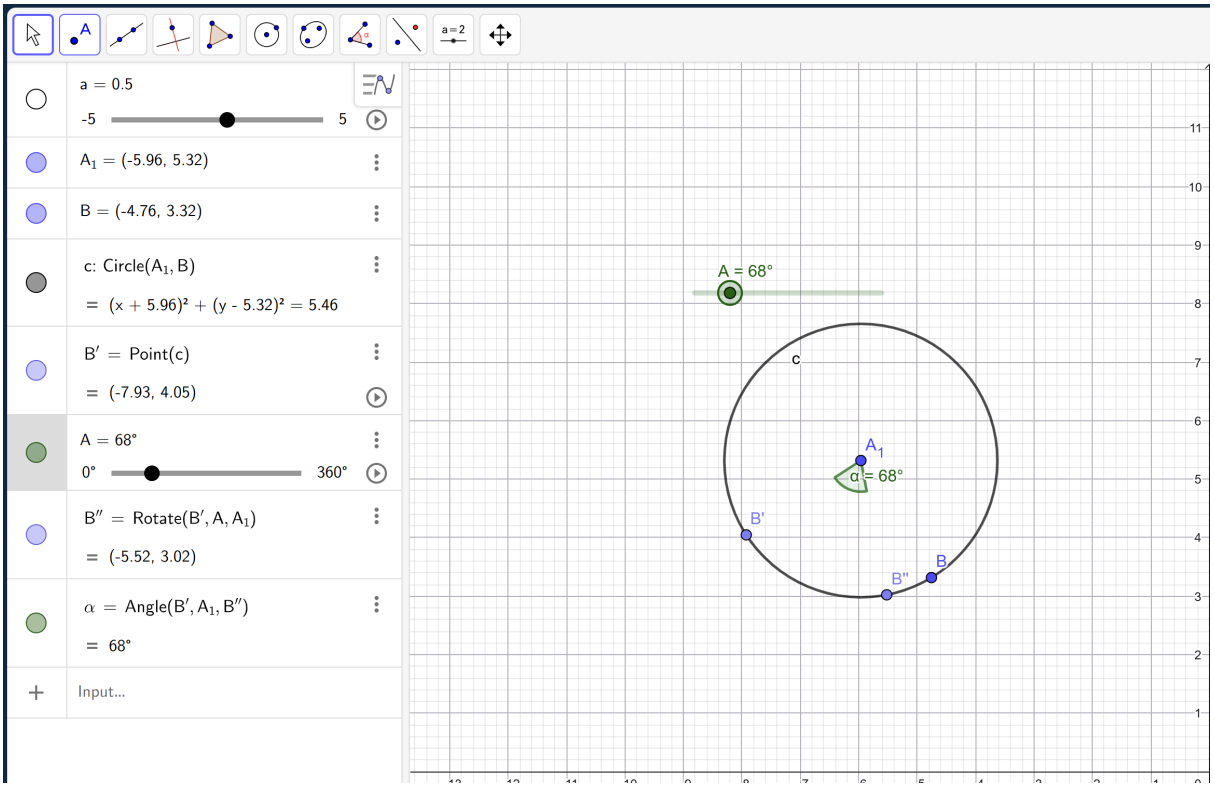


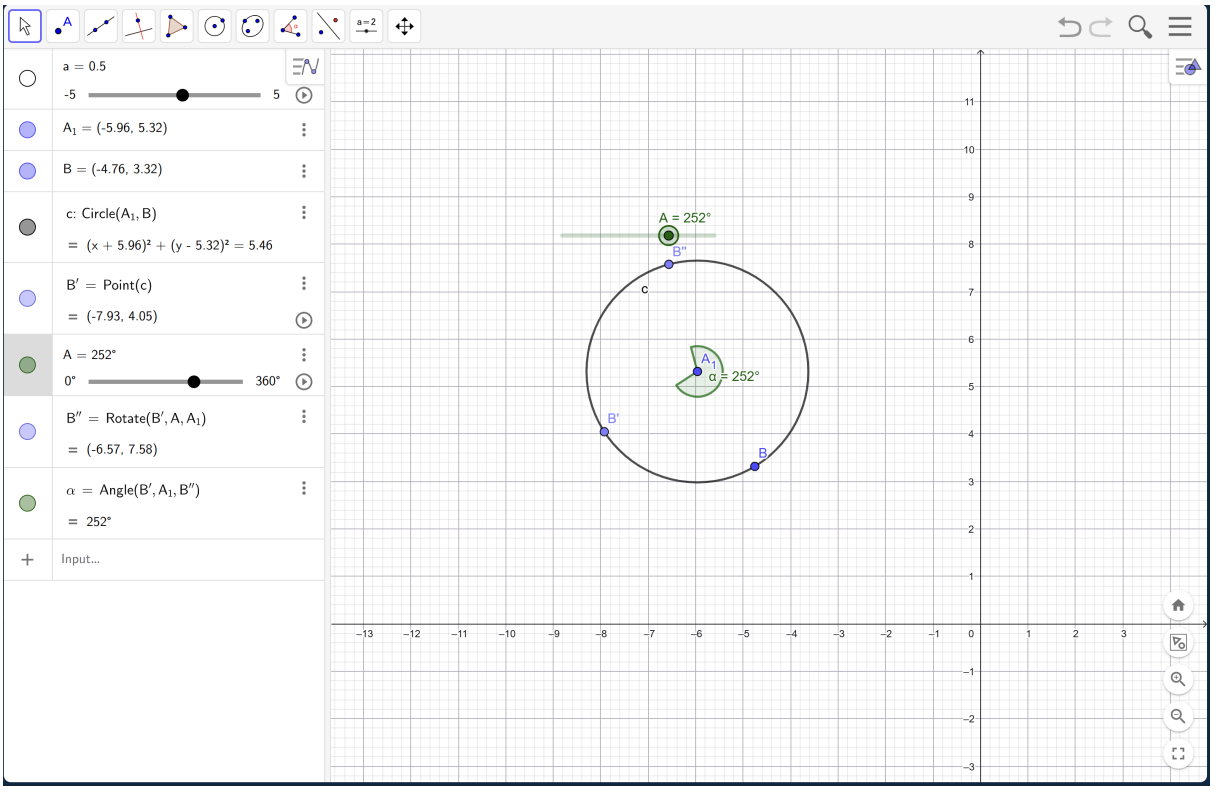
Performed the drag test and the center holds.

Task 3

Task 3 (10–12 min): Mini parametric model with a slider and associated chord/arc.

- Introduce a slider to control a parameter (such as an angle or length). Use this
  parameter to define a moving point dynamically rather than placing it freely.
- Create a chord or arc that depends on this moving point. When the slider changes, all
  related objects should update automatically.


a = 0.5
-5
5
A1 = (-5.96, 5.32)
B = (-4.76, 3.32)
c: Circle(A1, B)
= (x + 5.96)² + (y - 5.32)² = 5.46
B′ = Point(c)
= (-7.93, 4.05)
A = 68°
0°
360°
B″ = Rotate(B′, A, A1)
= (-5.52, 3.02)
α = Angle(B′, A1, B″)
= 68°
Input...
A = 68°
c
A1
α = 68°
B'
B''
B

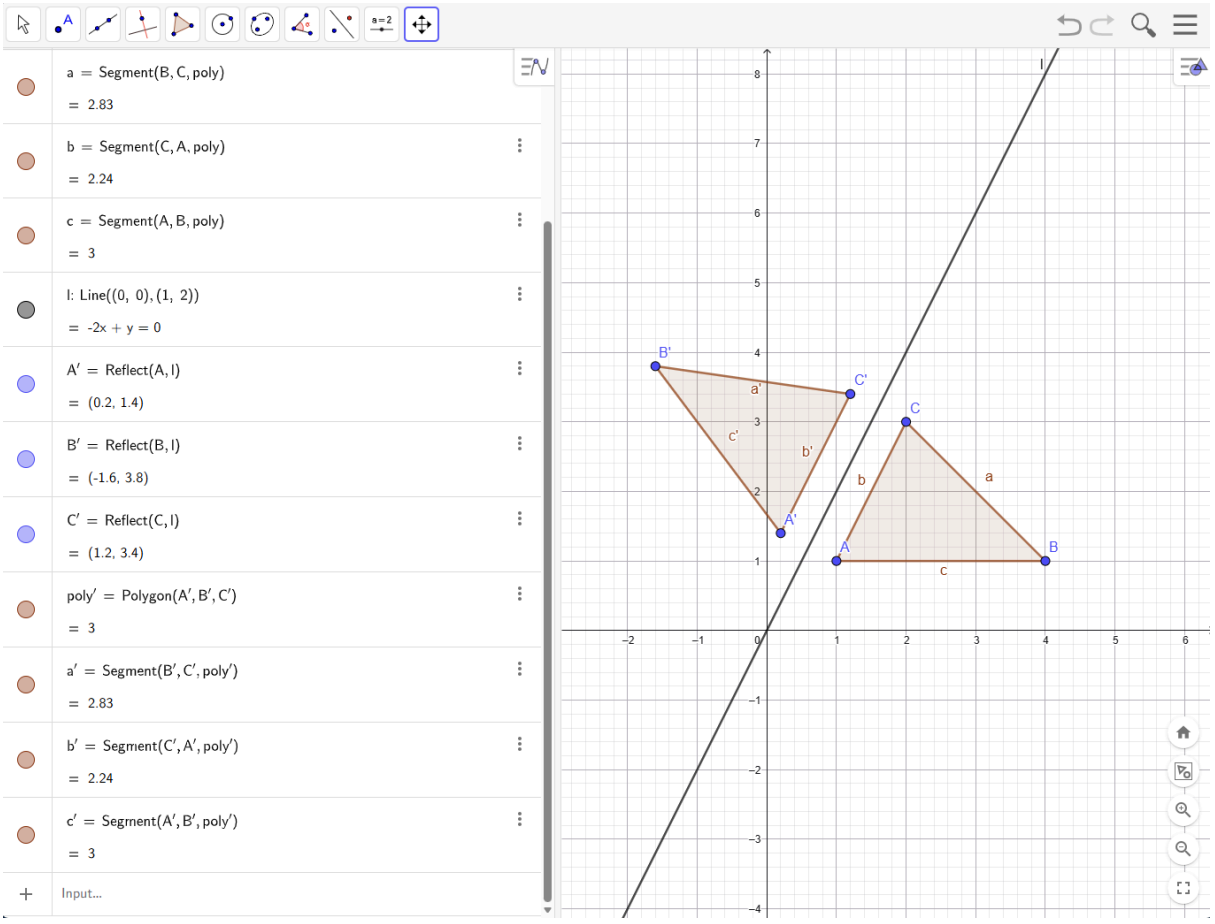

a = 0.5
-5
5
A1 = (-5.96, 5.32)
B = (-4.76, 3.32)
c: Circle(A1, B)
= (x + 5.96)² + (y - 5.32)² = 5.46
B′ = Point(c)
= (-7.93, 4.05)
A = 252°
0°
360°
B″ = Rotate(B′, A, A1)
= (-6.57, 7.58)
α = Angle(B′, A1, B″)
= 252°
Input...
A = 252°
B''
c
A1
α = 252°
B'
B

Task 4

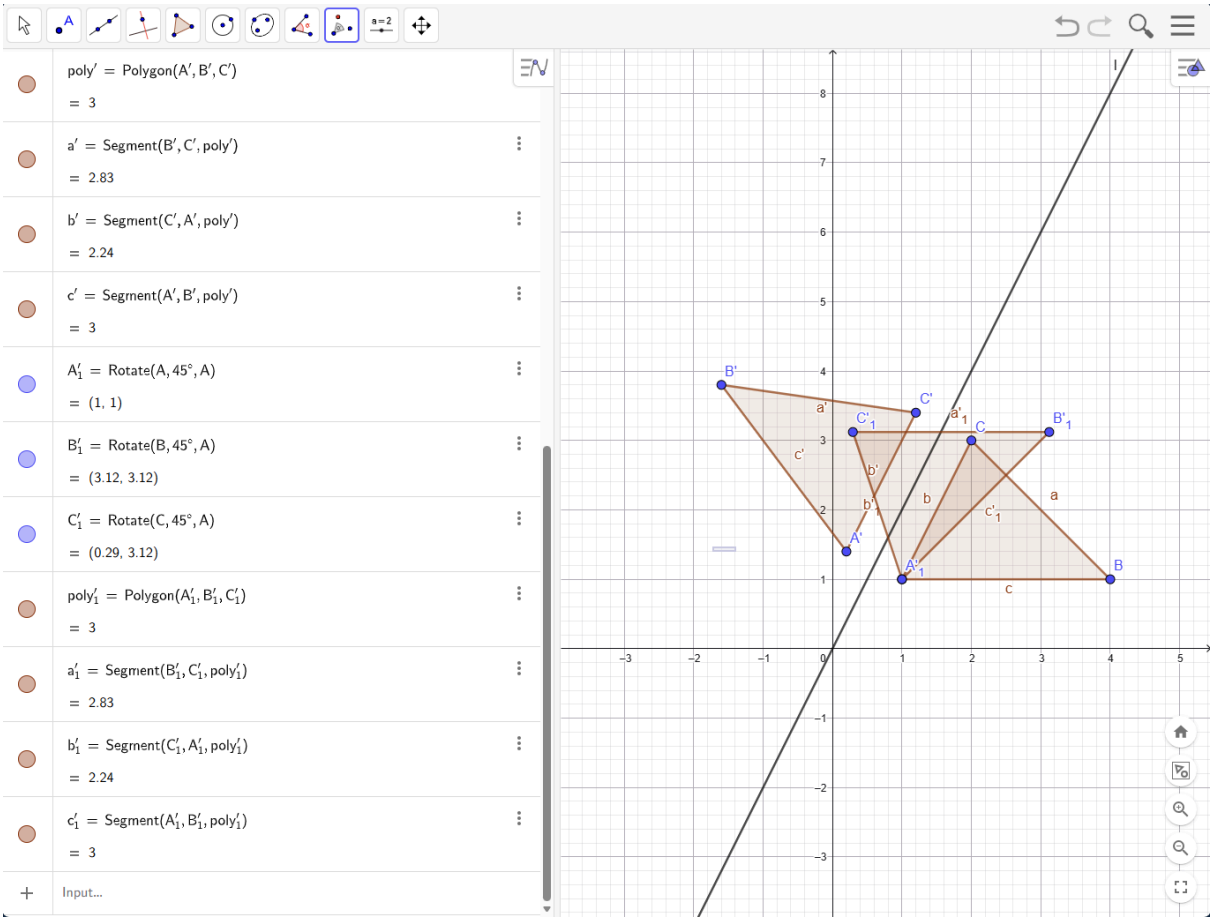
Task 4 (12–15 min): Composite transformations (reflect, rotate, dilate) and styling.

- Apply geometric transformations (reflection, rotation, dilation) using the transformation tools, not by redrawing shapes manually. Each transformed object
  should remain dependent on the original.
- Style the objects differently to distinguish them. Use the drag test to confirm that
  transformations remain valid when the original object is moved.

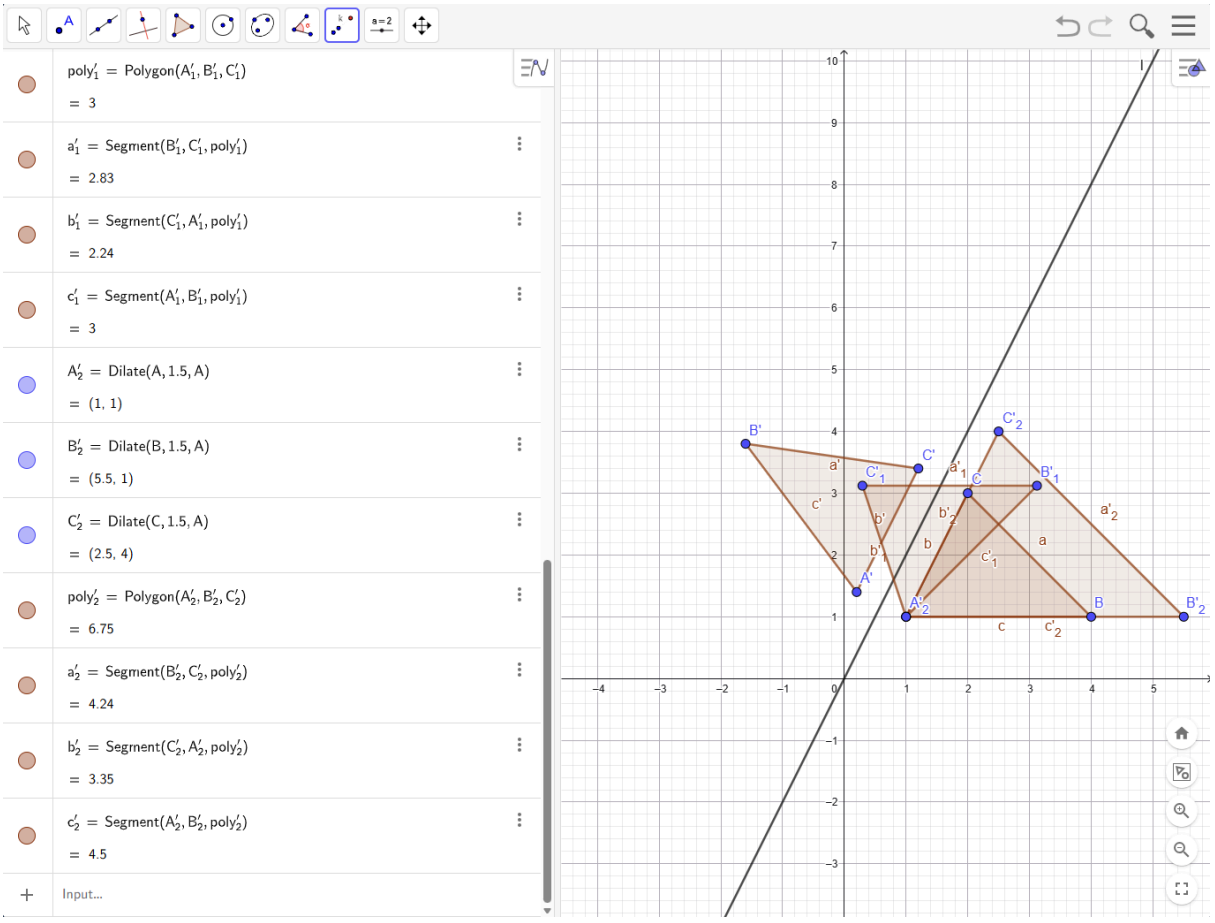
Reflection about a line l:



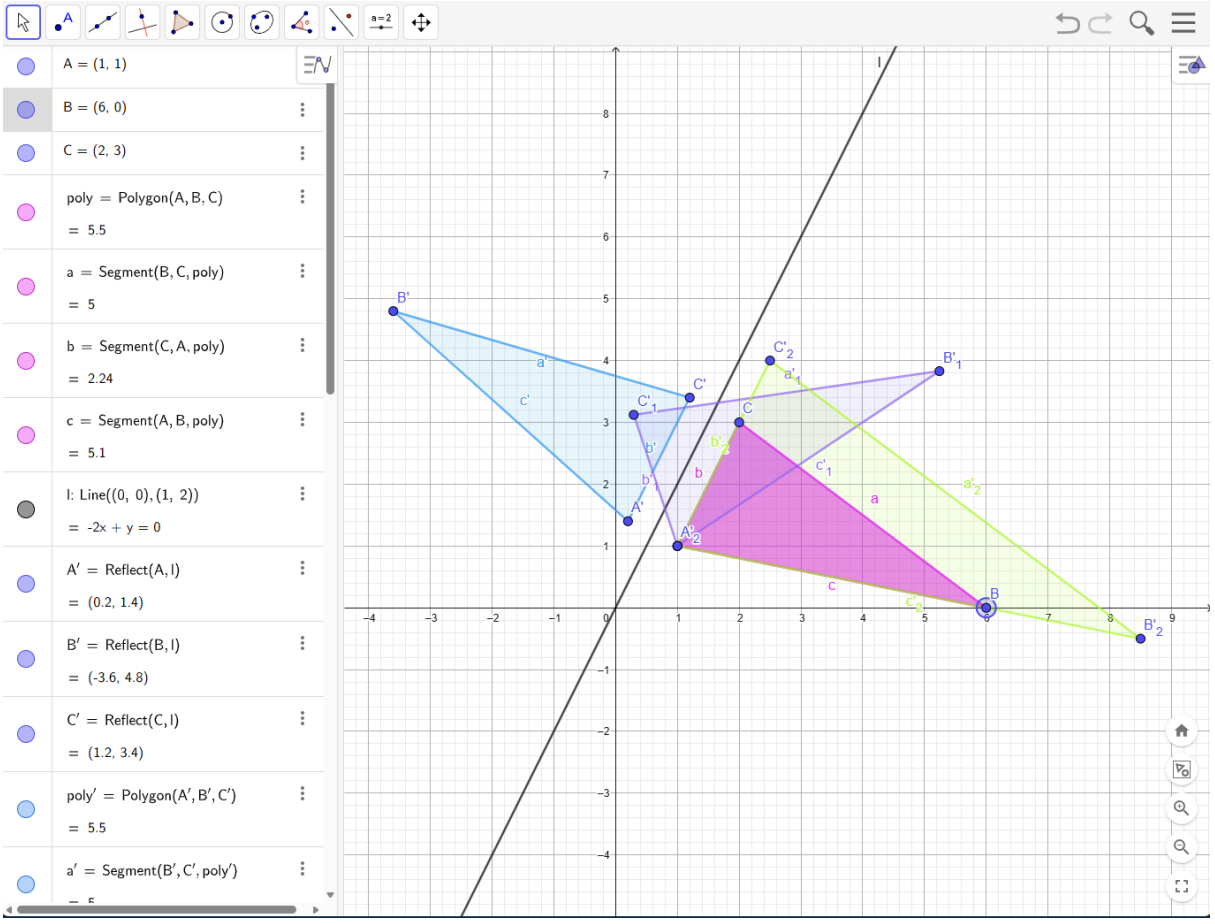
Rotation around point A with 45 degrees:



Dilation from Point A by factor of 1.5:



Styling the objects differently:



Performed the drag test (see image above) and everything holds up.

Task 1: https://www.geogebra.org/classic/cp4ubk3n

Task 2: https://www.geogebra.org/classic/ghbmqacn

Task 3: https://www.geogebra.org/classic/vzgcmqge

Task 4: https://www.geogebra.org/classic/hmjnfmmc